ECE 340: Semiconductor Devices

Lecture 15: Diffusion

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Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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Diffusion

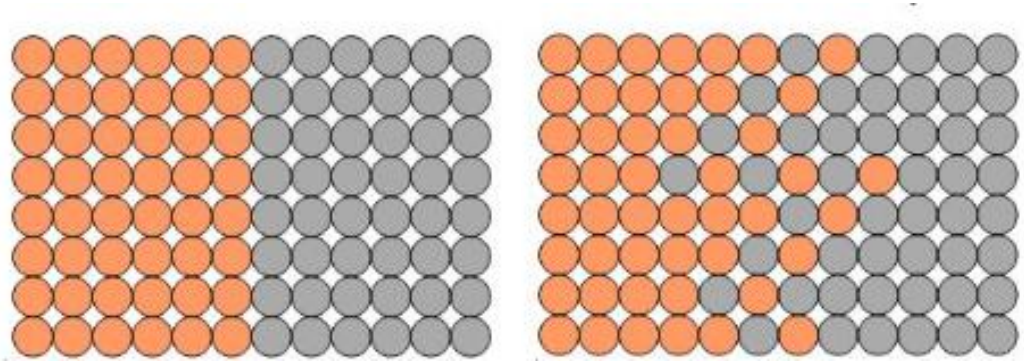
Diffusion

Stochastic process, from high concentration to low concentration region

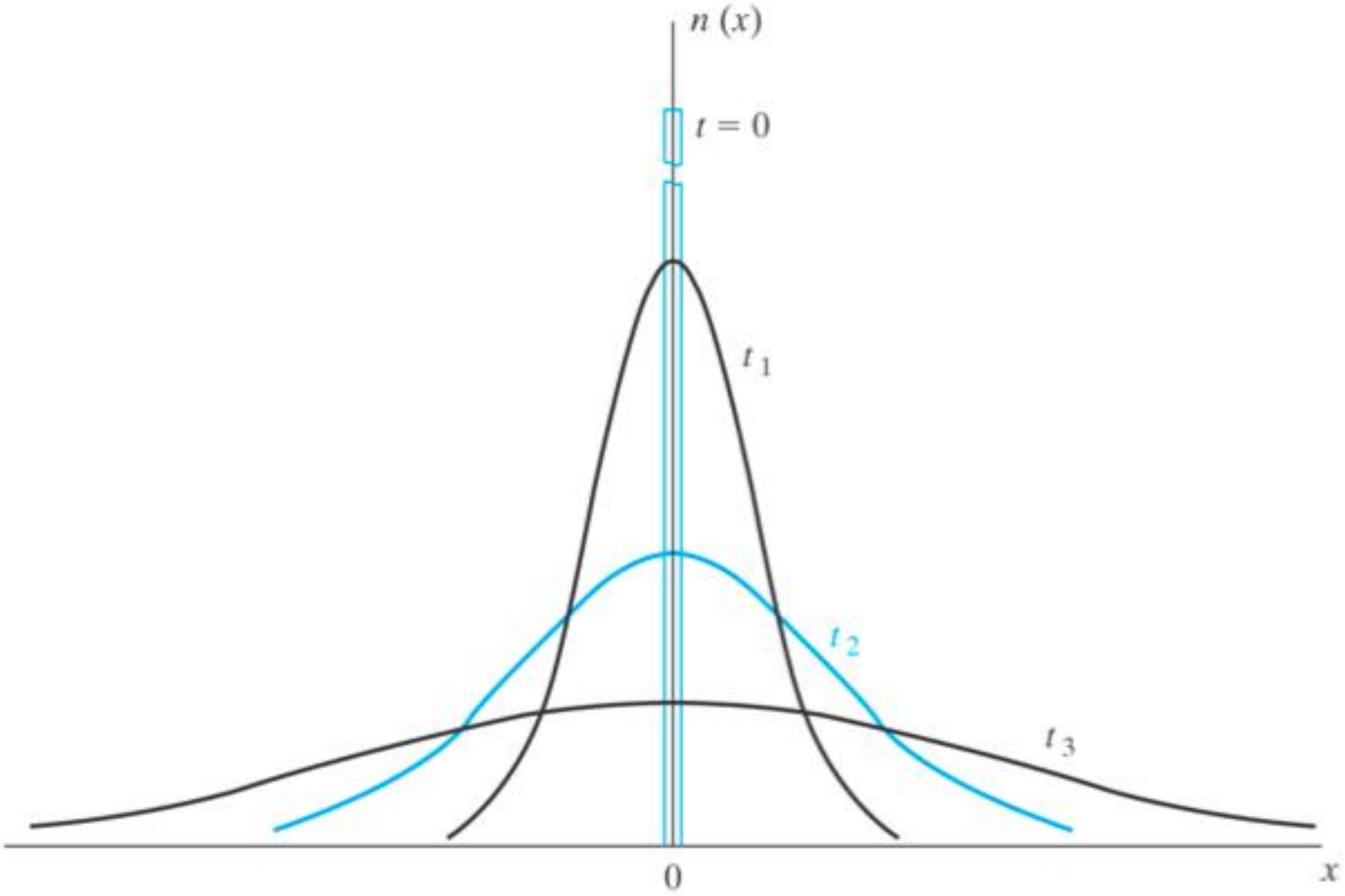
Diffusion takes place when the following conditions are present:

1. A step or gradient in distribution
2. Sufficient energy – usually thermal
3. A medium that supports random motion**

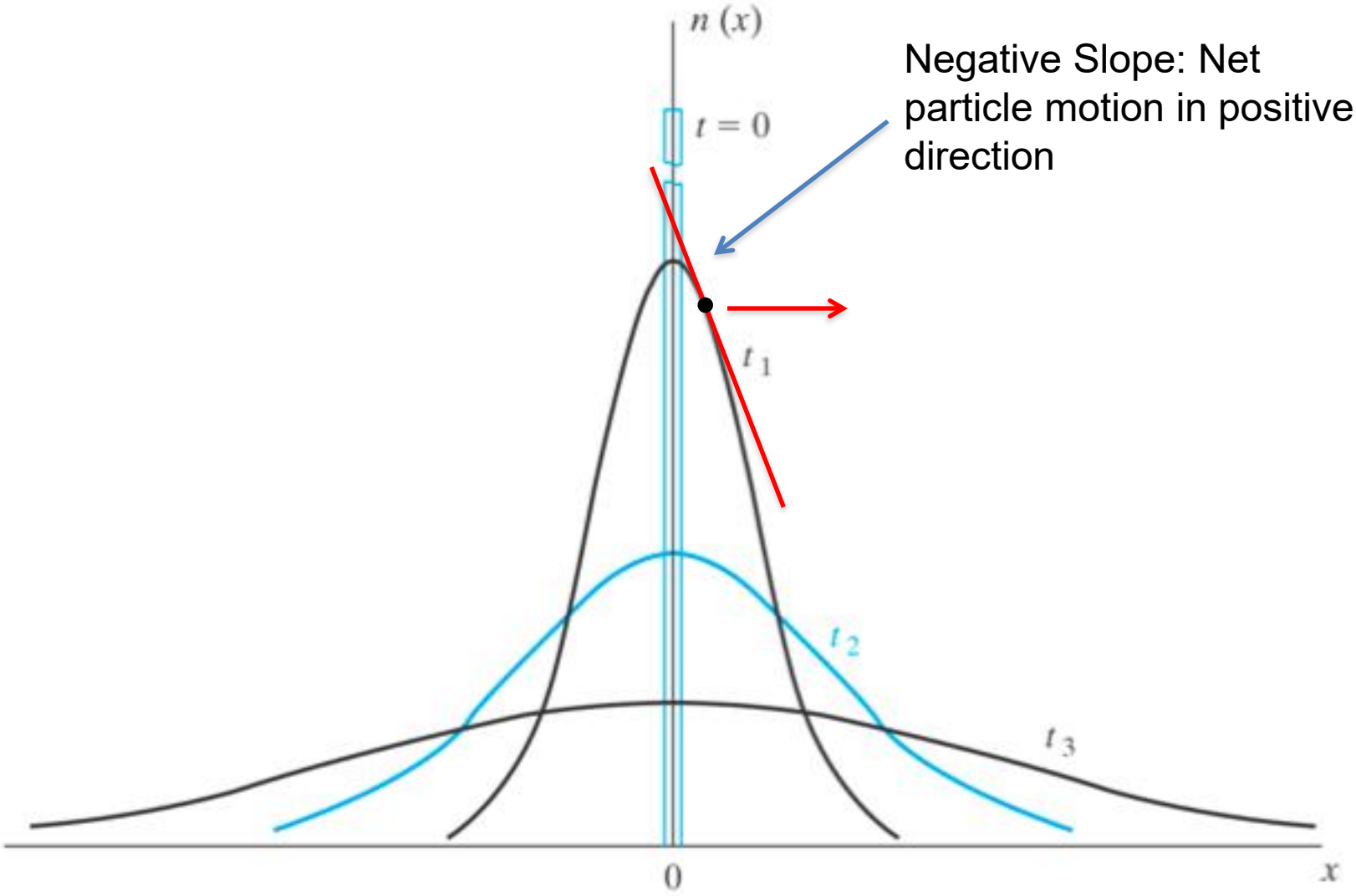




Diffusion

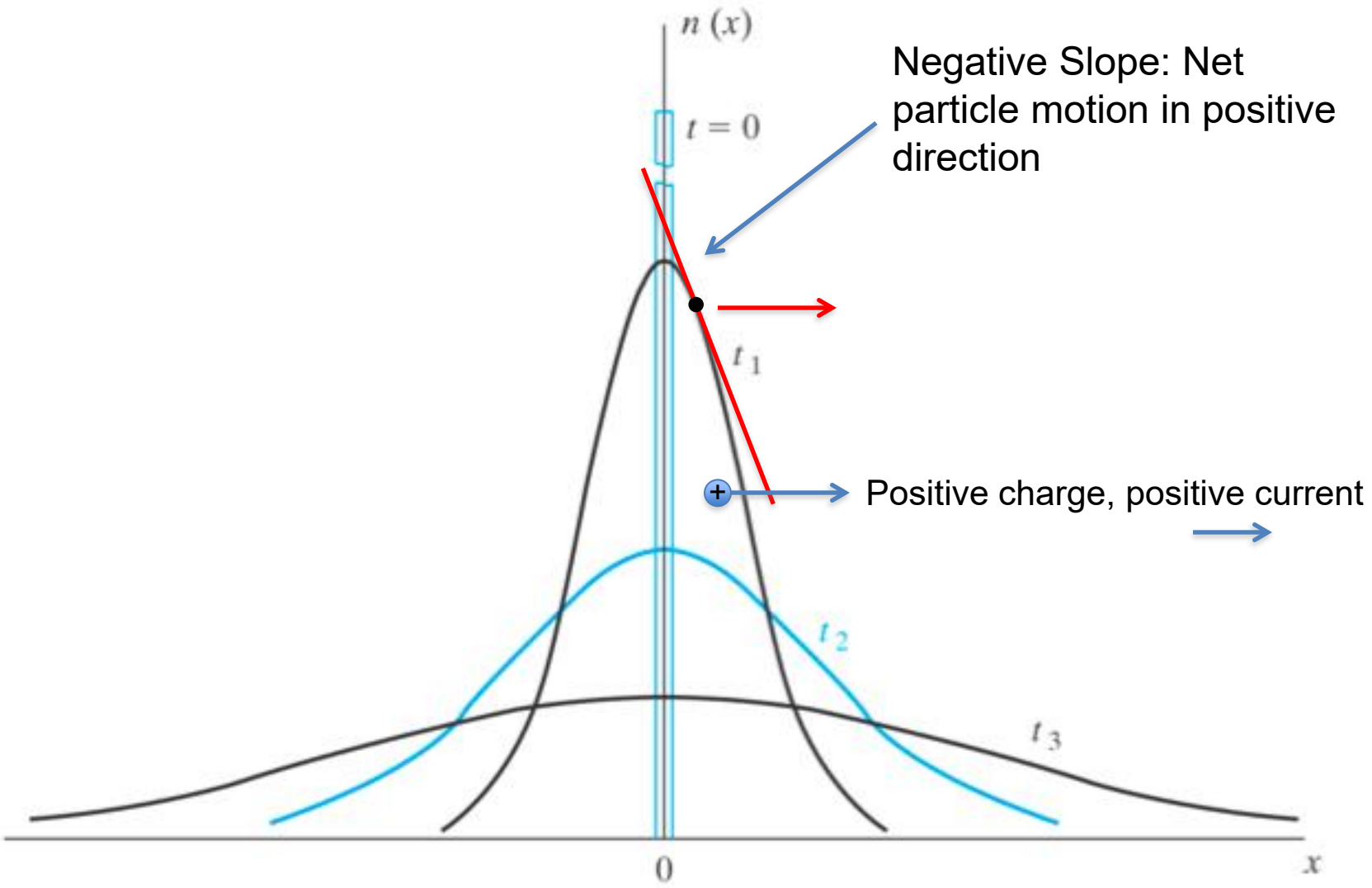


Diffusion: Net Motion & Slope

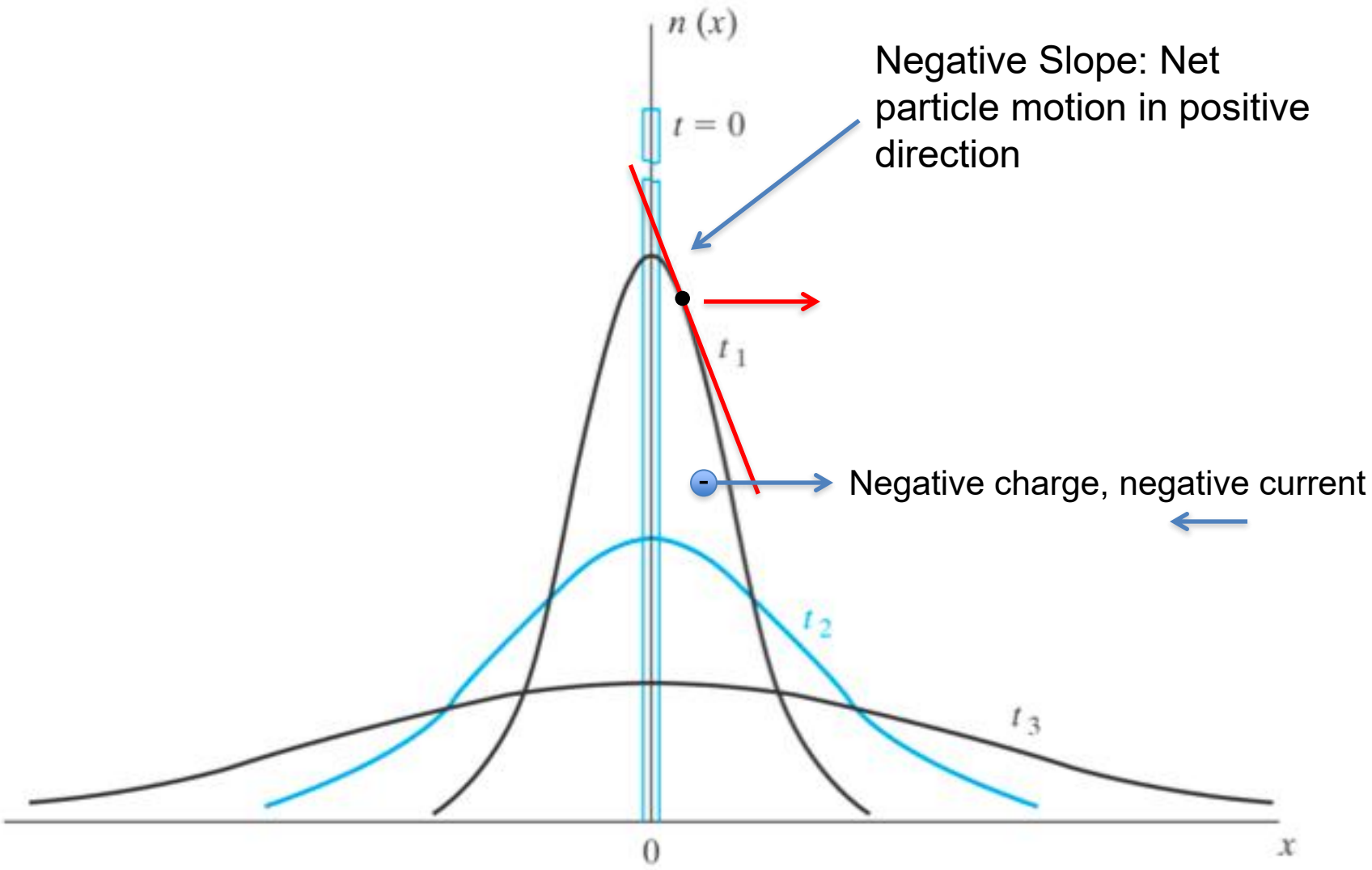


Negative Slope: Net particle motion in positive direction

Diffusion: Net Current

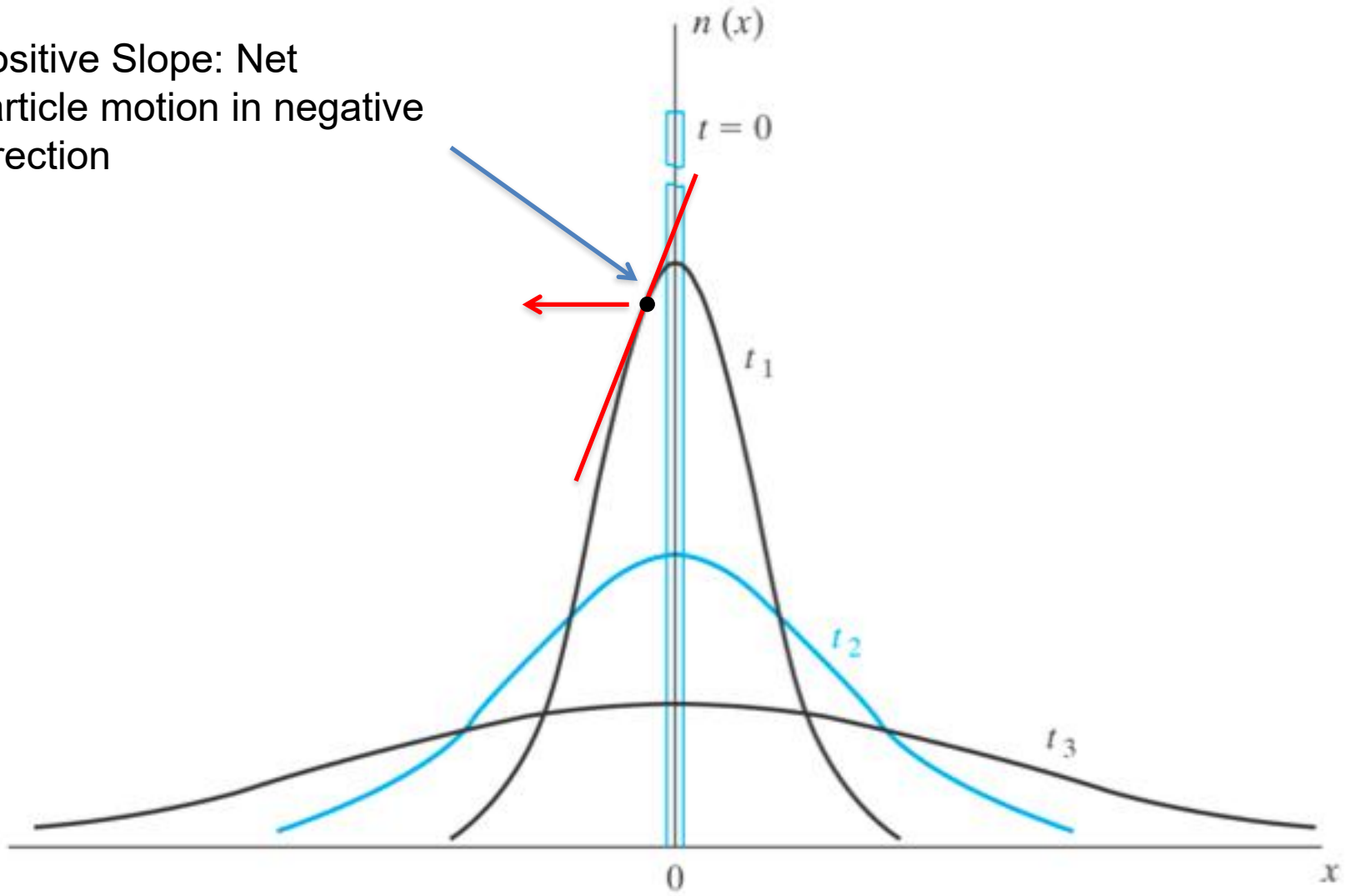


Diffusion: Net Current



Diffusion: Net Motion & Slope

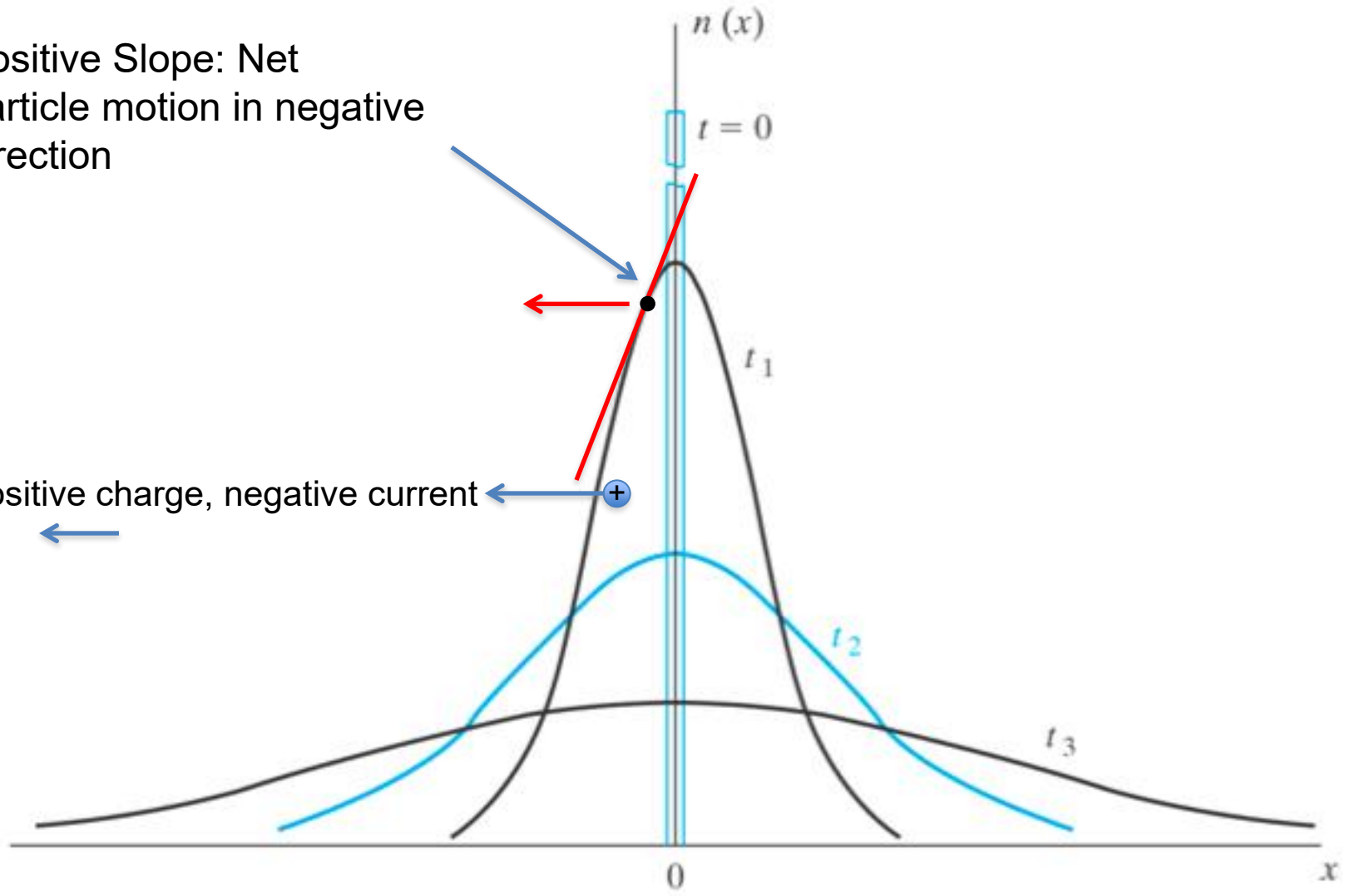
Positive Slope: Net particle motion in negative direction



Diffusion: Net Current

Positive Slope: Net particle motion in negative direction

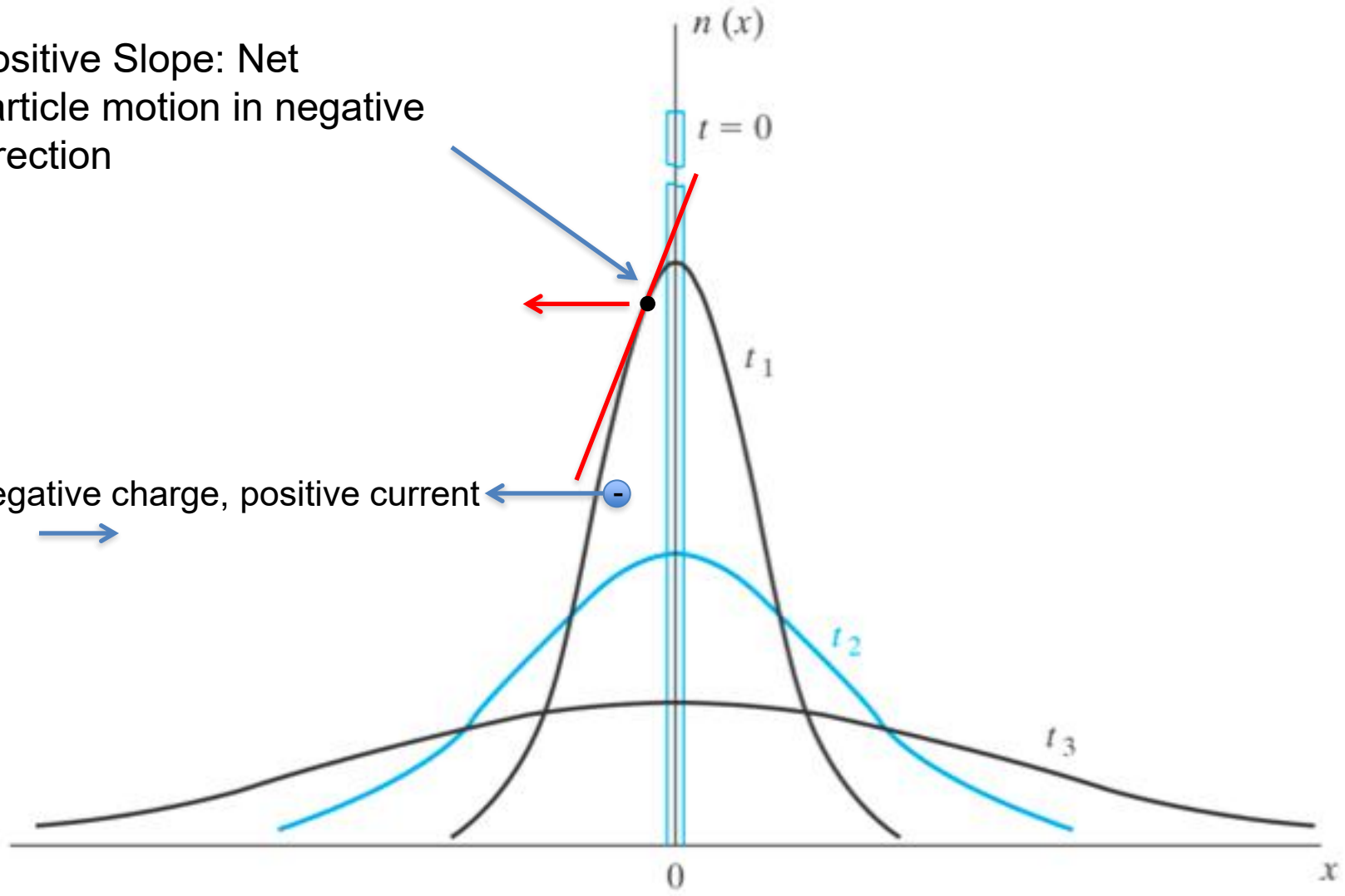
Positive charge, negative current



Diffusion: Net Current

Positive Slope: Net particle motion in negative direction

Negative charge, positive current



Diffusion Current

1) Divide distribution in segments $\bar{l}$ wide.

2) Evaluate n(x) at the center of each segment.

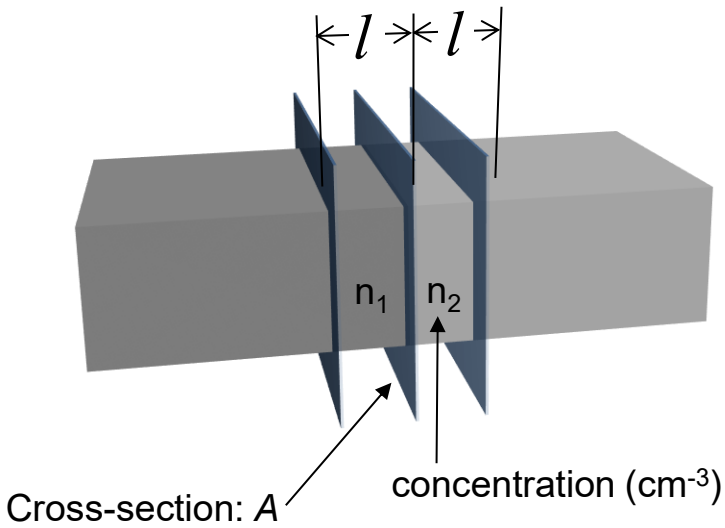
Consider the motion in a mean free time $\bar{t}$

where half move into each adjacent section.

The net number of electrons passing from left to right ($\rightarrow$) in time $\bar{t}$ is:

Net particle (electron) flow = $\frac{1}{2}\left(n_1 \bar{l} A\right) - \frac{1}{2}\left(n_2 \bar{l} A\right)$

A rate of electron flow can be defined: $\phi_n(x_o) = \frac{\bar{l}}{2\bar{t}}\left(n_1 - n_2\right)$



Note:

1) $\bar{l}$ is the mean free path between collisions

2) $\bar{t}$ is the mean free time for half the population to move one mean free path away

Diffusion Current

$$\phi_n(x_o) = \frac{\bar{l}}{2\bar{t}}(n_1 - n_2)$$

Since $\bar{l}$ is small: $n(x) - n(x+\Delta x) = (n_1 - n_2)\frac{\Delta x}{\bar{l}}$ and thus $(n_1 - n_2) = \frac{n(x) - n(x+\Delta x)}{\Delta x}\bar{l}$

In the limit of small Δx: $\phi_n(x) = \frac{\bar{l}}{2\bar{t}} \lim_{\Delta x \to 0}\left(\frac{n(x) - n(x+\Delta x)}{\Delta x}\right) = \frac{-\bar{l^2}}{2\bar{t}}\frac{dn(x)}{dx}$

The term $\frac{\bar{l^2}}{2\bar{t}}$ is the electron diffusion coefficient D_n with units cm^2/s

$$D_n = \frac{\bar{l^2}}{2\bar{t}}$$

A similar relation can be derived for hole motion. So:

$$\phi_n(x) = -D_n\frac{dn(x)}{dx}$$

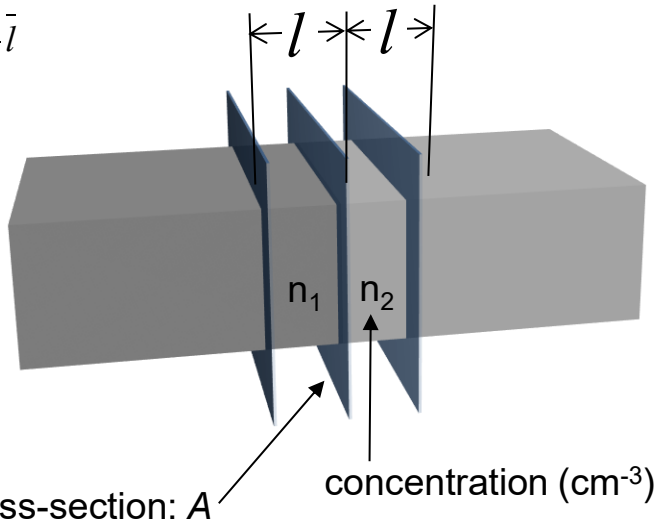
$$\phi_p(x) = -D_p\frac{dp(x)}{dx}$$

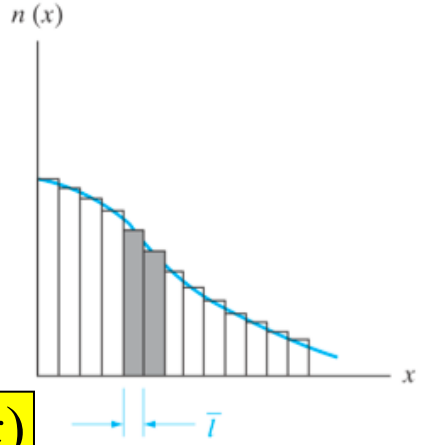
Electron and hole flux can be expressed as current:

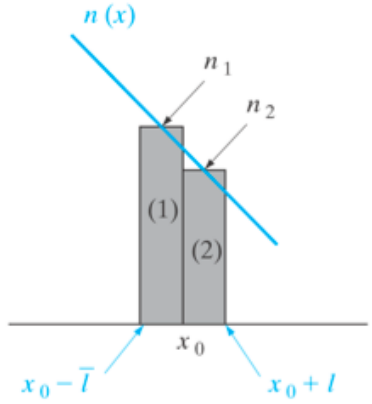
$$J_n(diff.) = -D_n\frac{dn(x)}{dx} \times (-q) = qD_n\frac{dn(x)}{dx}$$

$$J_p(diff.) = -D_p\frac{dp(x)}{dx} \times (+q) = -qD_p\frac{dp(x)}{dx}$$

$$J_n(x) = qD_n\frac{dn(x)}{dx} \text{ and } J_p(x) = -qD_p\frac{dp(x)}{dx}$$







Drift and Diffusion

Drift in a Semiconductor (Review)

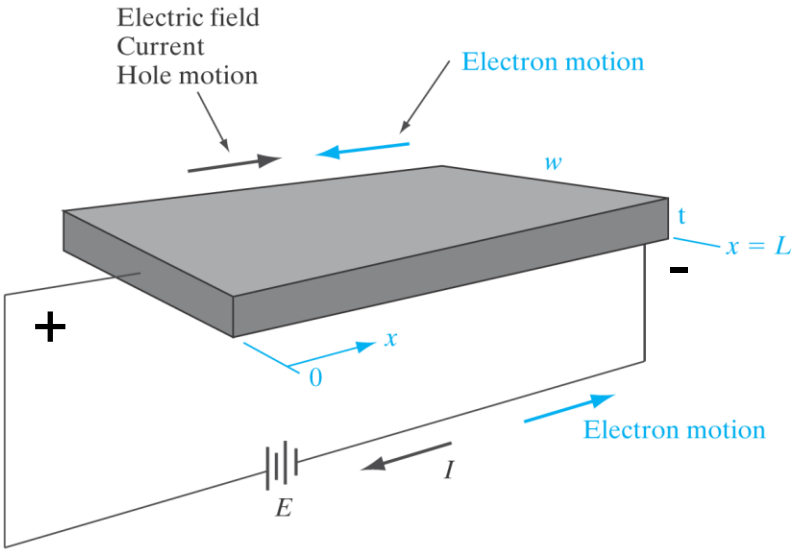


Figure 3.21

Drift of electrons and holes in a semiconductor bar.

Combined Current

$$J_{x,total} = q(n\mu_n + p\mu_p)E_x = \sigma E_x$$

$$\sigma = q(n\mu_n + p\mu_p)$$

$$R = \frac{\rho L}{A} = \frac{\rho L}{wt} = \frac{1}{\sigma}\frac{L}{wt} = \frac{1}{(n\mu_n + p\mu_p)}\frac{L}{wt}$$

Recall:

$$\langle v_e \rangle = -\mu_n E_x; \quad \langle v_p \rangle = \mu_p E_x$$

$$J_n(drift) = (-q)n\langle v_e \rangle = qn\mu_n E_x$$

$$J_p(drift) = (q)p\langle v_p \rangle = qp\mu_p E_x$$

Assume uniform distribution of carriers: no diffusion

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Particle Flux and Current Density

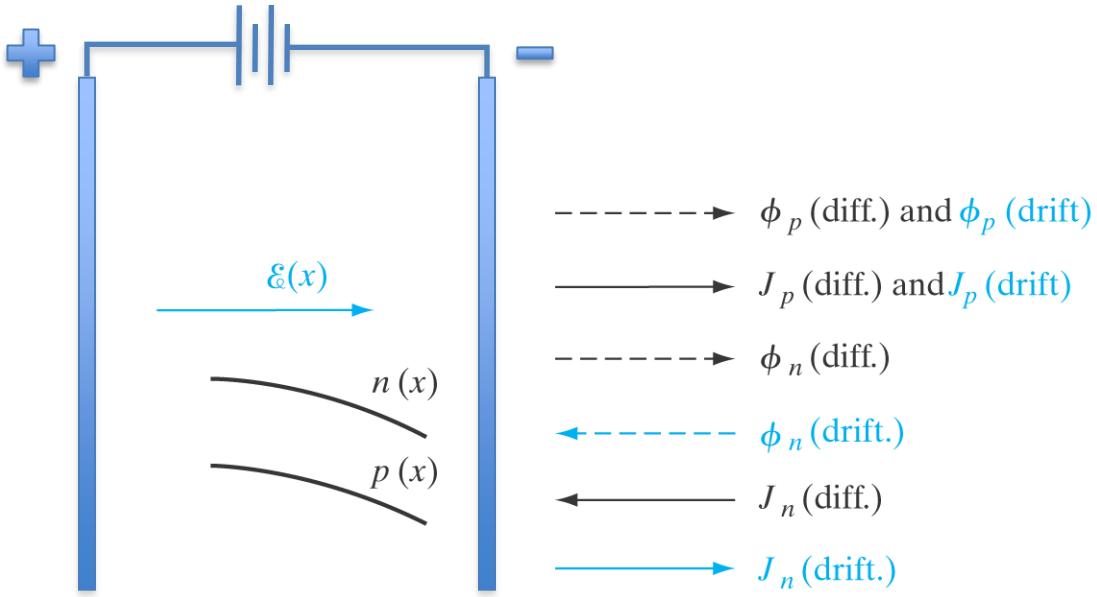


Figure 4.14

Drift and diffusion directions for electrons and holes in a carrier gradient and an electric field. Particle flow directions are indicated by dashed arrows, and the resulting currents are indicated by solid arrows.

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Total Current: Drift and Diffusion

$$J_n(x) = J_n(drift) + J_n(diff)$$

$$\rightarrow J_n(x) = qn(x)\mu_n\mathcal{E}(x) + qD_n\frac{dn(x)}{dx}$$

$$J_p(x) = J_p(drift) + J_p(diff)$$

$$\rightarrow J_p(x) = qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx}$$

$$\rightarrow J(x) = J_n(x) + J_p(x)$$

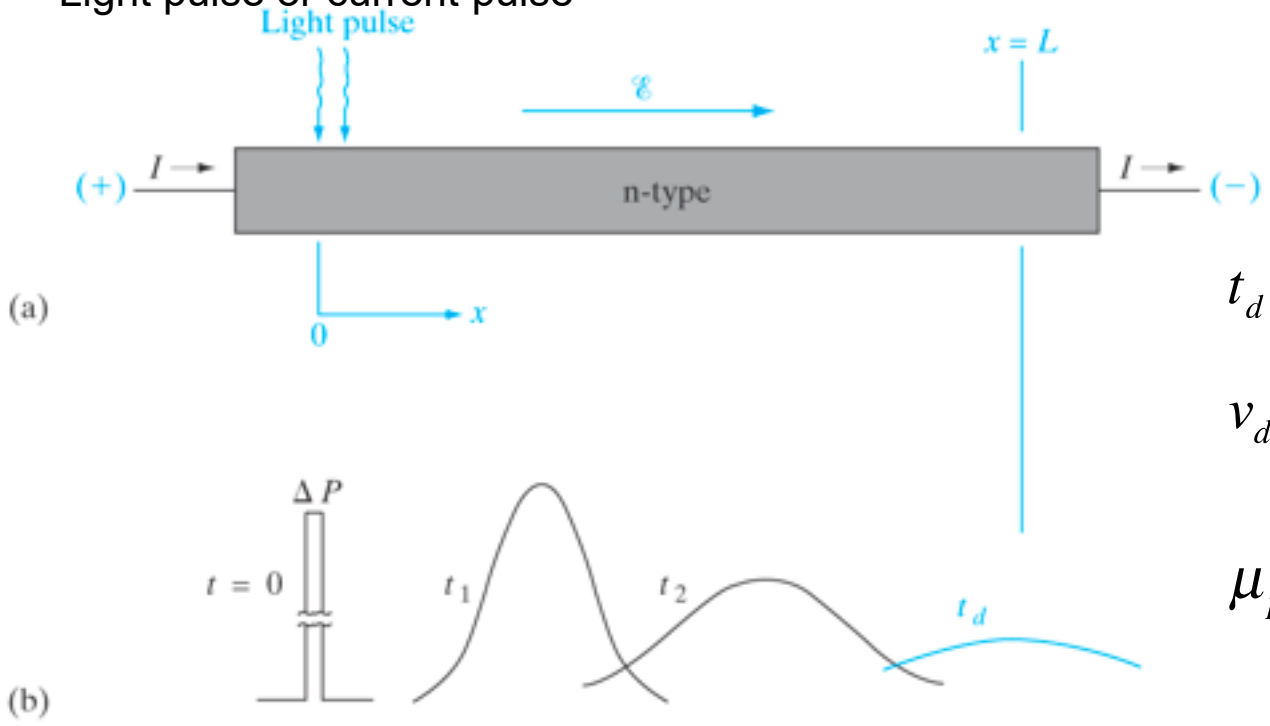
$$J(x) = qn(x)\mu_n\mathcal{E}(x) + qD_n\frac{dn(x)}{dx} + qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx}$$

The Haynes-Shockley Experiment

Bonus Material

Haynes-Shockley Experiment

Light pulse or current pulse



t_d = transit time

$$v_d = \frac{L}{t_d}$$

$$\mu_p = \frac{v_d}{\mathcal{E}} = \frac{v_d}{V/L}$$

Figure 4.18

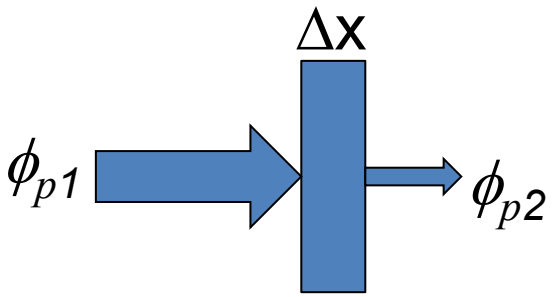
Drift and diffusion of a hole pulse in an n-type bar: (a) sample geometry; (b) position and shape of the pulse for several times during its drift down the bar.

Hall Effect: Majority carrier mobility
Haynes-Shockley: Minority carrier mobility

Haynes-Shockley Experiment

$$\phi_p(x) = -D_p \frac{dp(x)}{dx}$$

With Negligible Recombination: $\left(\frac{\delta p}{\tau_p}\right) = 0$, then by the law of conservation of matter



$$\frac{\Delta \delta p}{\Delta t} = \frac{-\Delta \phi_p}{\Delta x}$$

$$\Delta \phi_p = \phi_{p1} - \phi_{p2}$$

$$\frac{\partial \delta p(x,t)}{\partial t} = D_p \frac{\partial^2 \delta p(x,t)}{\partial x^2}$$

Haynes-Shockley Experiment

With Negligible Recombination: $\left(\frac{\delta p}{\tau_p}\right) = 0$

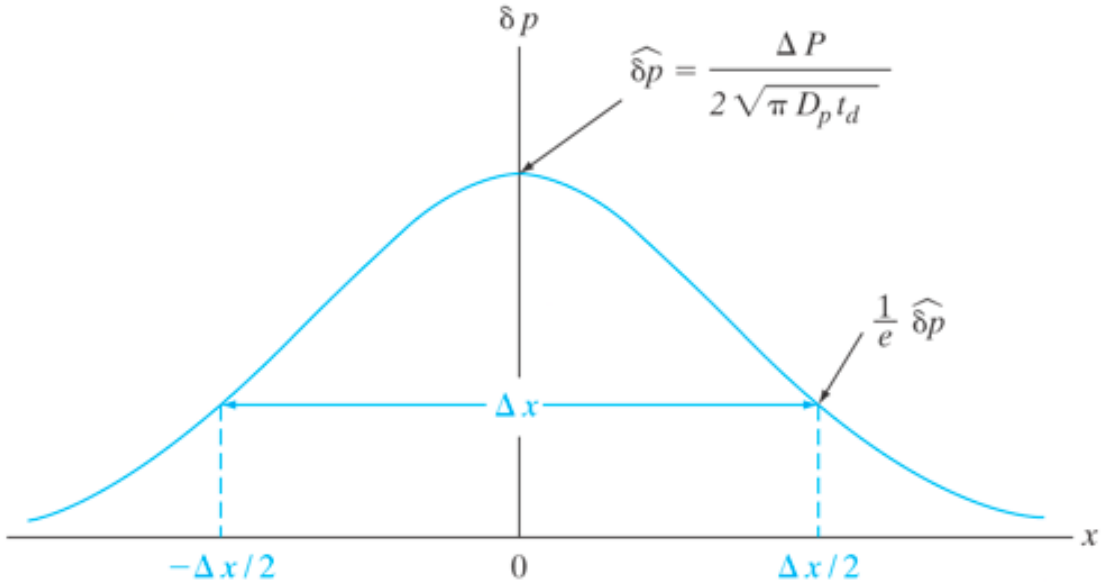
$$\frac{\partial \delta p(x,t)}{\partial t} = D_p \frac{\partial^2 \delta p(x,t)}{\partial x^2}$$ ← Diffusion Equation

Solution:

$$\delta p(x,t) = \left[\frac{\Delta P}{2\sqrt{\pi D_p t}}\right] e^{-\frac{x^2}{4D_p t}}$$

$$\delta \hat{p} = \frac{\Delta p}{2\sqrt{\pi D_p t}}$$ ← Amplitude of Peak

$e^{-\frac{x^2}{4D_p t}}$ ← Spread of Pulse in +/-x Directions



At $\frac{1}{e}$ Point:

$$\delta \hat{p} \cdot e^{-1} = \delta \hat{p} \cdot e^{-\frac{\left(\Delta x/2\right)^2}{4D_p t_d}}$$

$$\Delta x = 4\sqrt{D_p t_d} = 4\sqrt{D_p \frac{L}{v_d}} = 4\sqrt{\frac{D_p L}{\mu E}}$$

$$\Delta x = \Delta t \cdot v_d = \Delta t \cdot \frac{L}{t_d}$$ ← Δt is more easily measured

$$D_p = \frac{(\Delta x)^2}{16 t_d} = \frac{\Delta t^2 L^2}{16 t_d^{\ 3}}$$

Haynes-Shockley Experiment

$$\Delta x = \Delta t \cdot v_d = \Delta t \cdot \frac{L}{t_d}$$

$$D_p = \frac{(\Delta x)^2}{16 t_d} = \frac{\Delta t^2 L^2}{16 t_d^{\,3}}$$

Since $v_d = \mu_p \mathcal{E} = \mu_p \frac{V}{L_o} = \frac{L}{t_d}$

$$\mu_p = \frac{L \cdot L_o}{V \cdot t_d}$$

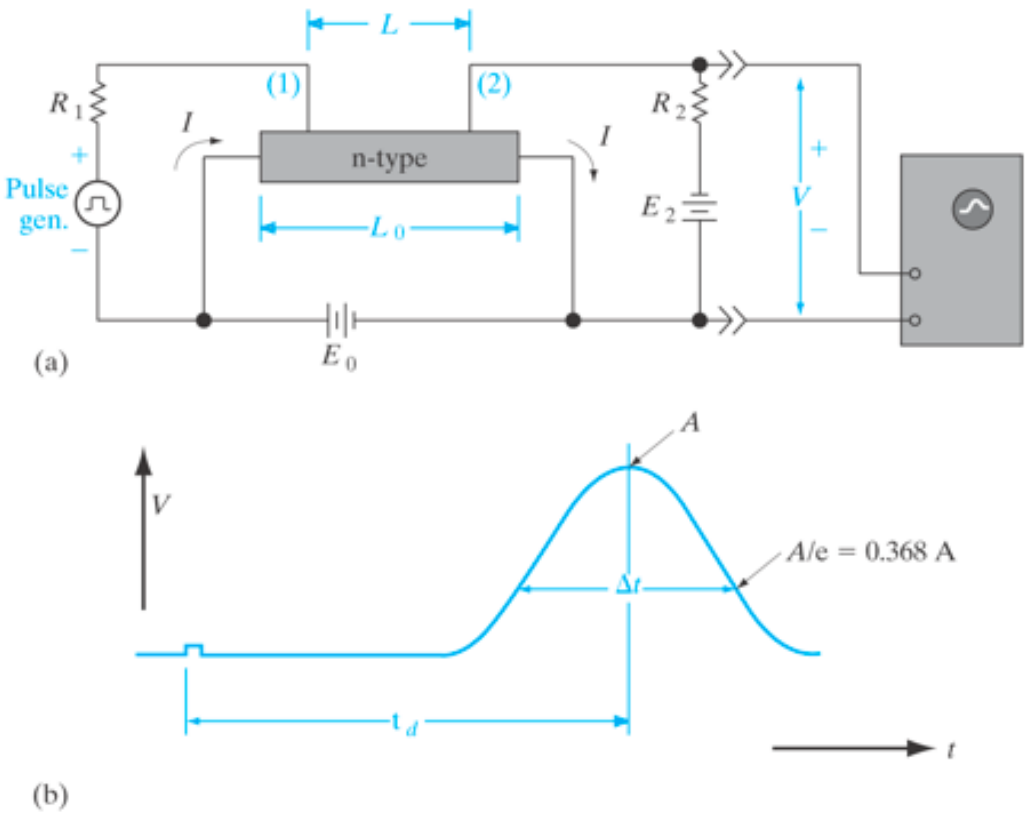


Figure 4.20

The Haynes–Shockley experiment: (a) circuit schematic; (b) typical trace on the oscilloscope screen.

Haynes-Shockley Experiment

Example:
Ge, Sample length 1cm
Probes separated by 0.95 cm
Voltage of 2V
Pulse arrives at t_d=0.25 ms
Pulse width Δt=117μs

$$D_p = \frac{(\Delta x)^2}{16 t_d} = \frac{\Delta t^2 L^2}{16 t_d^3} = \frac{\left(117x10^{-6}s\right)^2 \cdot (0.95cm)^2}{16 \cdot (0.25x10^{-3}s)^3} = 49.4 cm^2/s$$

$$\mu_p = \frac{v_d}{\mathcal{E}} = \frac{\frac{L}{t_d}}{\frac{E_o}{L_o}} = \frac{\frac{0.95cm}{0.25x10^{-3}s}}{\frac{2V}{1cm}} = 1900 cm^2/V-s$$

$$\frac{D_p}{\mu_p} = 0.026V = \frac{kT}{q}$$

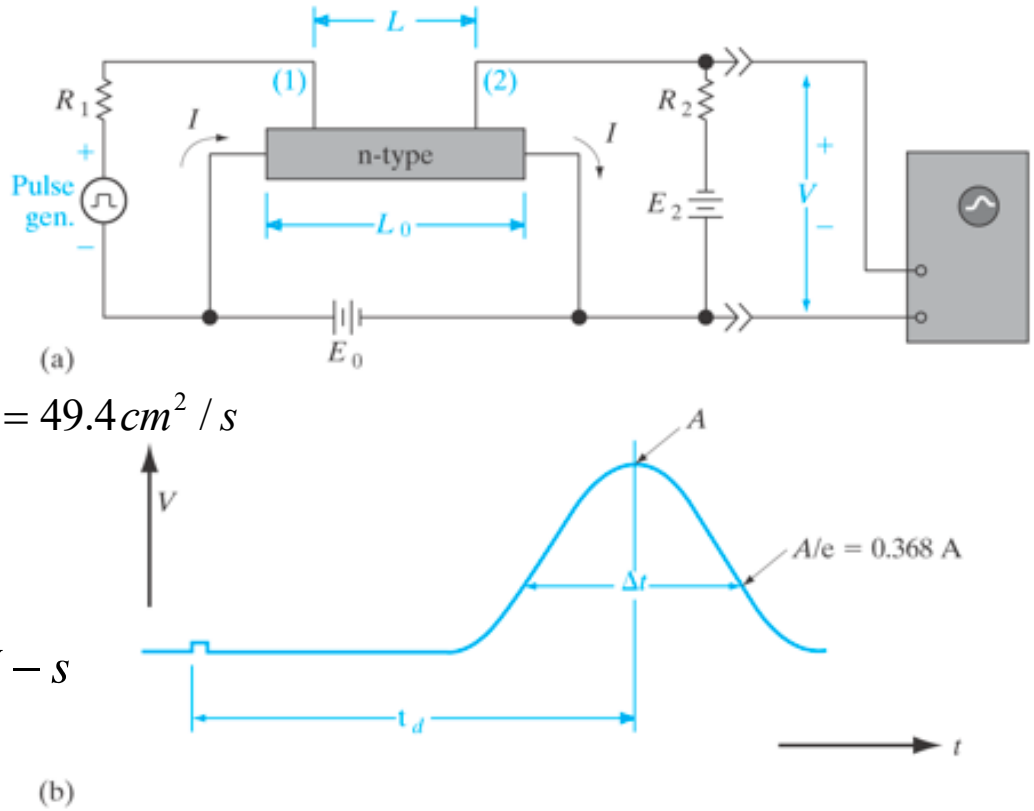


Figure 4.20

The Haynes–Shockley experiment: (a) circuit schematic; (b) typical trace on the oscilloscope screen.